Replicating a Experimental Study (THE I)

Research Design and Methods in Quantitative Research

Dr. K. Jonathan Klüser

2026-10-15

Instructions

Please read and follow the guidelines below carefully. Then, complete the exercises and report the results in a Quarto document. Compile the Quarto document in PDF and submit both the compiled PDF and `.qmd` files within the deadline.

Further instructions about the submission are below.

Preparation step 1: Install R and RStudio

To complete this exercise, you will need **R** and **RStudio**. If you don't have them installed already, download and install them from:

- R
- RStudio

A tutorial on how to start using R and R Studio is here. Please **contact the lecturer** and collaborate with your classmates if you have any doubts or need help.

Preparation step 2: Prepare a Quarto Document

Open RStudio, create a new Quarto document (`.qmd`), and set the output format to PDF. Make sure your Quarto installation is up-to-date:

```
# Install Quarto if needed
## Run this line in a separate script or the Quarto document will not compile
# install.packages("quarto")
```

You can find help on how to set up a Quarto document here.

Preparation step 3: Read the Assigned Paper and Download the Replication Files

You will need to read the paper *Instrumentally Inclusive: The Political Psychology of Homonationalism*, which you can find in the exercise materials folder.

After reading the paper, download the Study 1 replication materials, also provided in the materials folder:

- Data file: `study1_data.csv`
- R script: `study1.R`

These files have been downloaded from the authors' replication repository. Take a moment to explore it—next time, you may need to download the files yourself!

If you look at their repository, you will notice that the replication materials **do not include a codebook** for the data, so you will need to navigate their R script to identify the relevant variables.

Exercises

Exercise 1: Summary of the Paper and Main Findings

Provide a brief summary of the paper and describe its main findings, especially those related to Study 1. Remember to be concise—two or three paragraphs are usually enough for a good summary.

Exercise 2: Data Exploration

Load the data file `study1_data.csv` into R and describe the dataset. Then, describe the experimental variables `treatment` and `imm_1` (i.e., pre-treatment attitudes toward immigration) by providing visualizations of their distributions. Next, select up to four covariates (e.g., gender, age, etc.) and plot their distributions as well. What do these visualizations tell us about the data?

Importantly, you may need to clean (e.g., recode or remove missing values) or transform (e.g., convert a numeric into a categorical variable) some of the variables to produce appropriate visualizations. Document any changes made.

```r
# Load necessary packages
library(tidyverse) # tidyverse environment
library(ggplot2) # nice plots

# Load the data
data <- read_csv("take_home_exercise_1_materials/study1_data.csv")

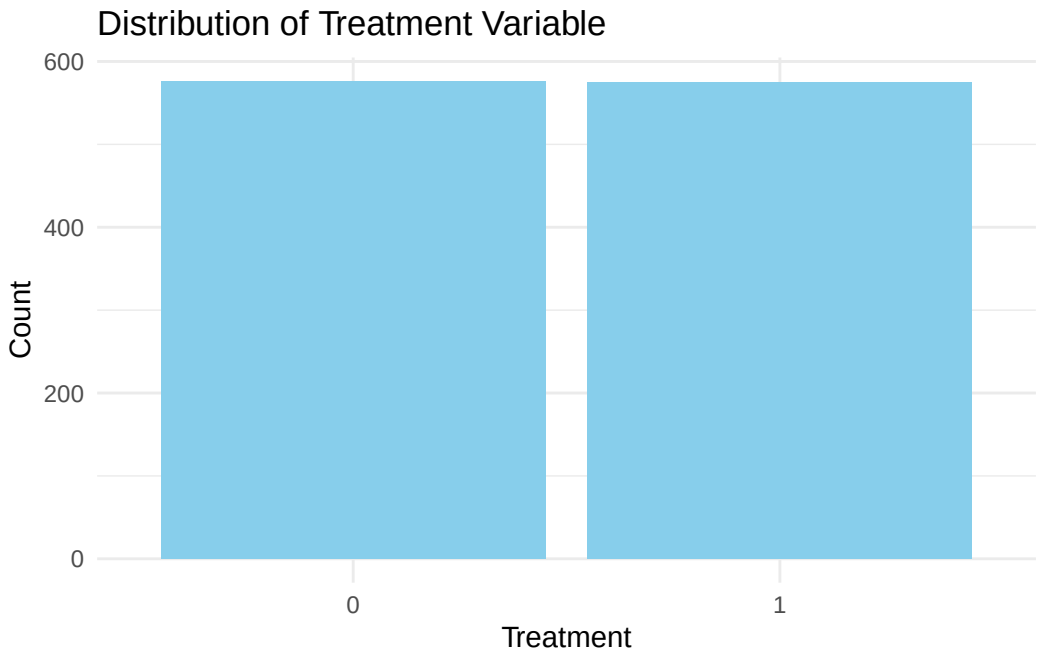
# Subset the data
data_subset <- data |>
  dplyr::select(
         support, # outcome variable
         treatment, # treatment variable
         imm_1, # conditional variable
         age,
         gender,
         degree,
         nonwhite
         ) |>
  # Factorizing variables
  mutate(treatment_fct = as.factor(treatment),
         gender_fct = as.factor(gender),
         gender = as.numeric(gender_fct),
         degree_fct = as.factor(degree),
         nonwhite_fct = as.factor(nonwhite),
         nonwhite = as.numeric(nonwhite_fct))

# Display summary of main variables
summary(data_subset)
```

```
    support         treatment          imm_1             age
 Min.   :0.0000   Min.   :0.0000   Min.   : 0.000   Min.   :18.00
 1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.: 4.000   1st Qu.:35.00
 Median :1.0000   Median :0.0000   Median : 7.000   Median :48.00
 Mean   :0.6533   Mean   :0.4996   Mean   : 6.099   Mean   :47.52
 3rd Qu.:1.0000   3rd Qu.:1.0000   3rd Qu.: 9.000   3rd Qu.:62.00
 Max.   :1.0000   Max.   :1.0000   Max.   :10.000   Max.   :88.00
                                                    NA's   :3
     gender          degree          nonwhite     treatment_fct gender_fct
 Min.   :1.000   Min.   :0.0000   Min.   :1.000   0:576         Man  :549
 1st Qu.:1.000   1st Qu.:0.0000   1st Qu.:1.000   1:575         Woman:599
 Median :2.000   Median :0.0000   Median :1.000                 NA's :  3
 Mean   :1.522   Mean   :0.4294   Mean   :1.169
```

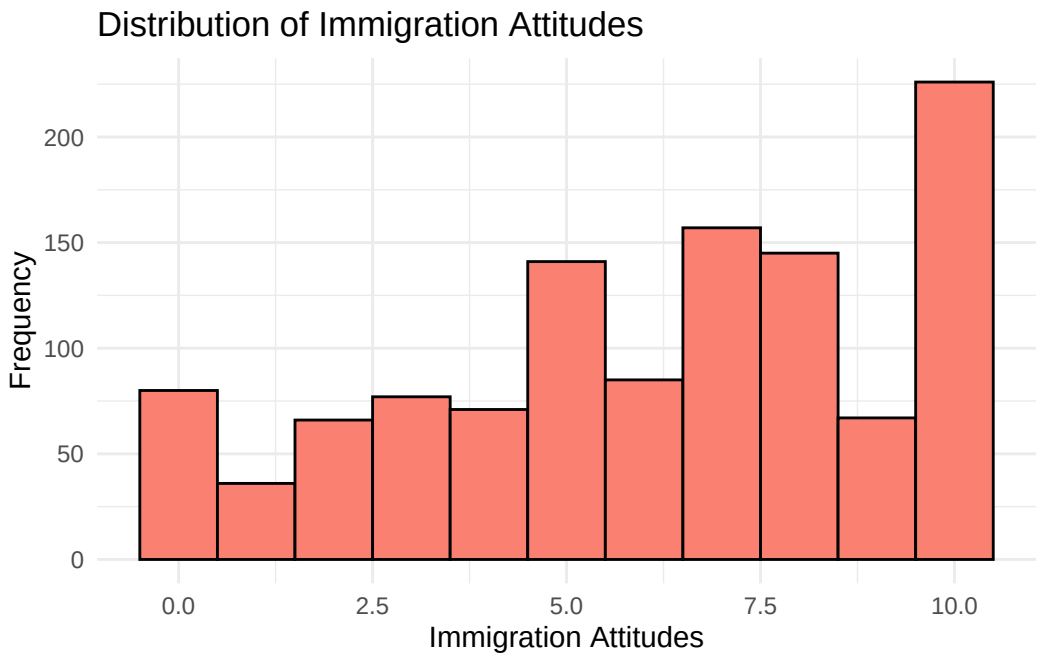
```
3rd Qu.:2.000   3rd Qu.:1.0000   3rd Qu.:1.000
Max.   :2.000   Max.   :1.0000   Max.   :2.000
NA's   :3       NA's   :3
degree_fct nonwhite_fct
0   :655   0:956
1   :493   1:195
NA's:  3
```

```r
# Visualize the distribution of the treatment variable
ggplot(data_subset, aes(x = treatment_fct)) +
  geom_bar(fill = "skyblue") +
  labs(title = "Distribution of Treatment Variable",
       x = "Treatment",
       y = "Count") +
  theme_minimal()
```



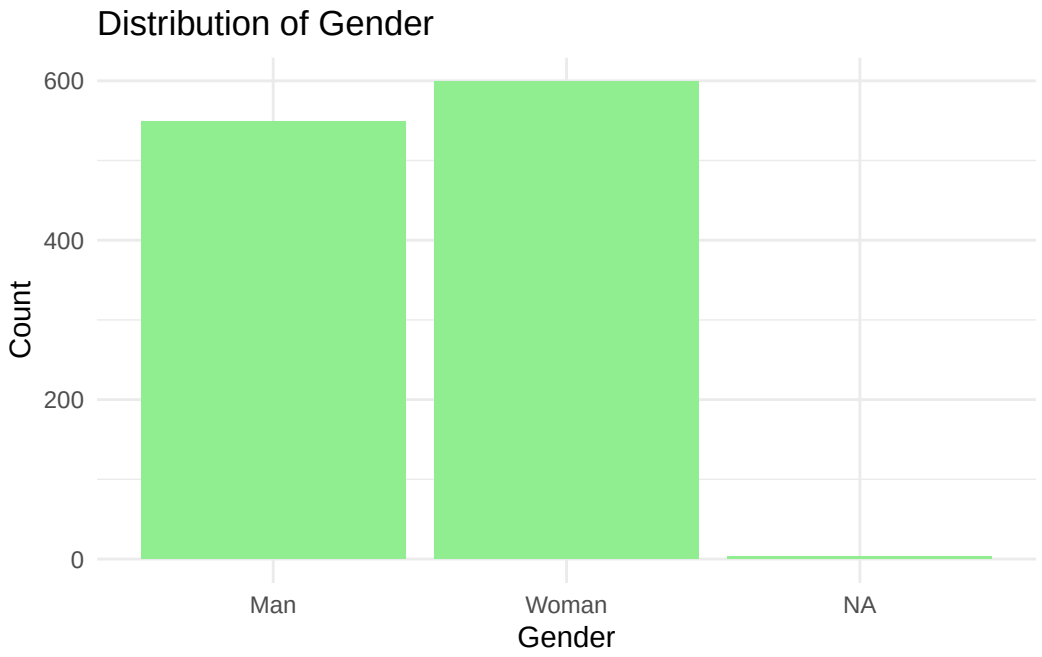

```r
# Visualize the distribution of the immigration attitudes variable
ggplot(data_subset, aes(x = imm_1)) +
```

```
  geom_histogram(binwidth = 1, fill = "salmon", color = "black") +
  labs(title = "Distribution of Immigration Attitudes",
       x = "Immigration Attitudes",
       y = "Frequency") +
  theme_minimal()
```

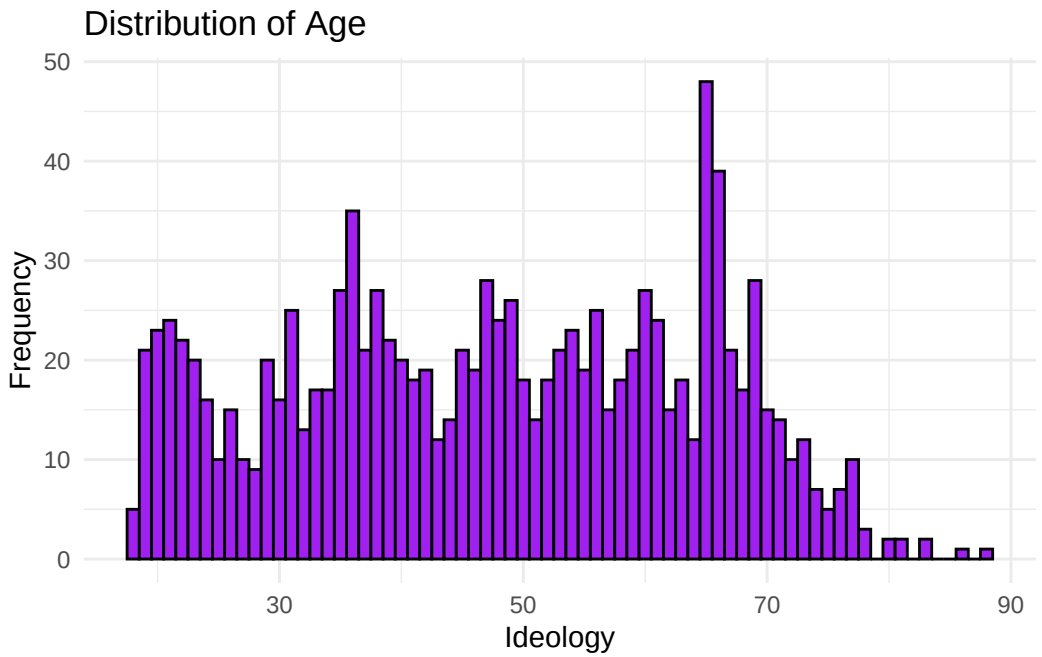



```
# Select and visualize distributions for up to four covariates

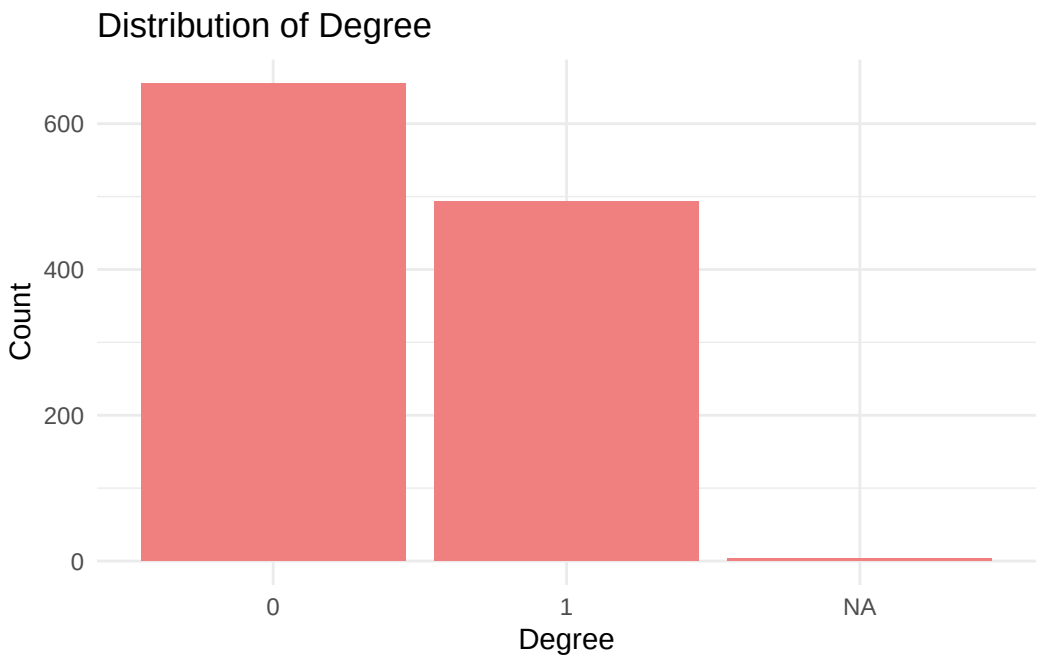
## Gender
ggplot(data_subset, aes(x = gender_fct)) +
  geom_bar(fill = "lightgreen") +
  labs(title = "Distribution of Gender",
       x = "Gender",
       y = "Count") +
  theme_minimal()
```



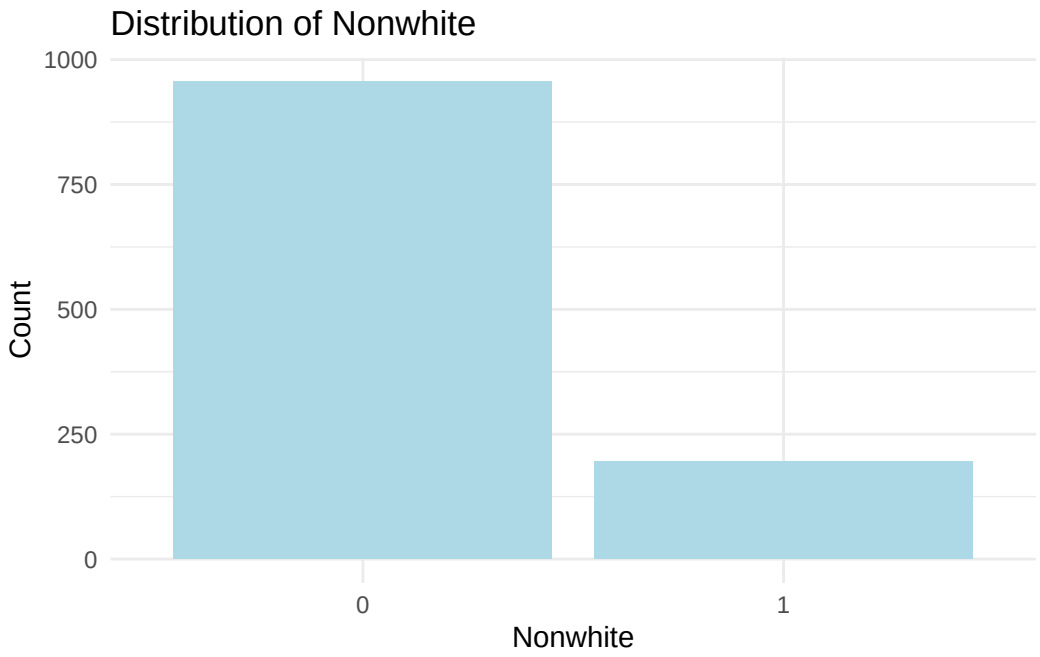

```
## Age
ggplot(data_subset, aes(x = age)) +
  geom_histogram(binwidth = 1, fill = "purple", color = "black") +
  labs(title = "Distribution of Age",
       x = "Ideology",
       y = "Frequency") +
  theme_minimal()
```




```
## Degree
ggplot(data_subset, aes(x = degree_fct)) +
  geom_bar(fill = "lightcoral") +
  labs(title = "Distribution of Degree",
       x = "Degree",
       y = "Count") +
  theme_minimal()
```




```
## Nonwhite
ggplot(data_subset, aes(x = nonwhite_fct)) +
  geom_bar(fill = "lightblue") +
  labs(title = "Distribution of Nonwhite",
       x = "Nonwhite",
       y = "Count") +
  theme_minimal()
```



Exercise 3: Check Covariate Balance

Check for balance across covariates by examining whether the mean values and distributions of your selected covariates are similar between the treatment and control groups. Report the results in a table *and* with plots.

Explain your findings. Why would you expect randomization to lead to balance across covariates?

```
# Load necessary packages
library(tableone) # tables for covariate balance
library(broom) # tidy data from models
library(purrr) # + tidy data from models

# Define covariates to check for balance
covariates <- c("imm_1", "age", "gender", "degree", "nonwhite")

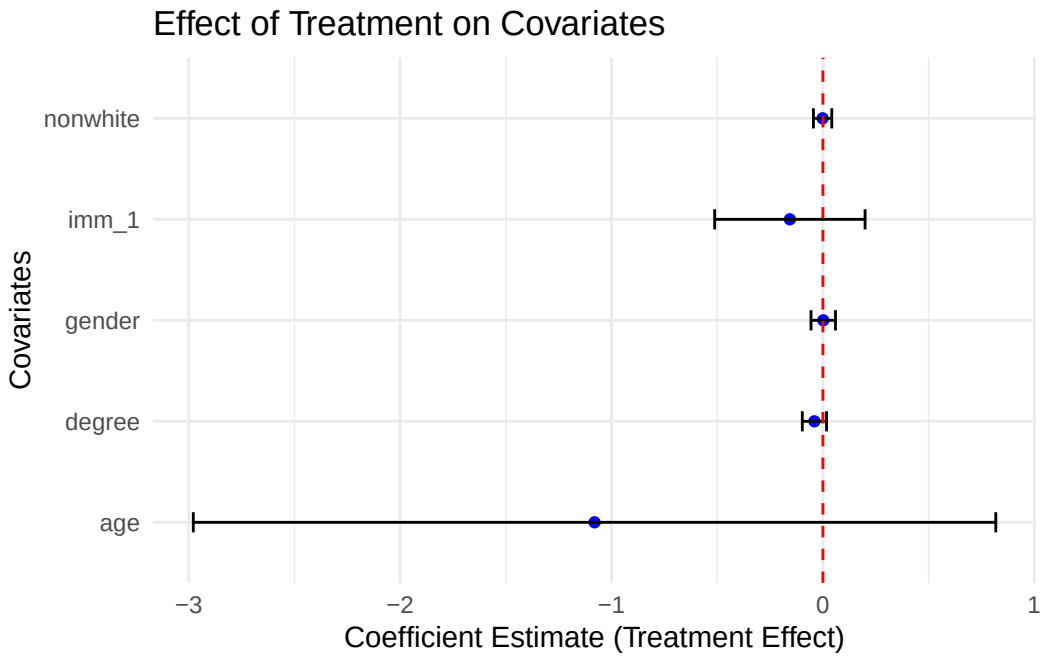
# Create a balance table by treatment
balance_table <- CreateTableOne(vars = covariates,
                                strata = "treatment_fct",
                                data = data_subset)
```

```r
# Print the balance table
print(balance_table, showAllLevels = TRUE)
```

```
                      Stratified by treatment_fct
                       level 0             1             p      test
  n                           576           575
  imm_1 (mean (SD))          6.18 (3.06)   6.02 (3.10)   0.390
  age (mean (SD))           48.06 (15.95) 46.98 (16.85)  0.265
  gender (mean (SD))         1.52 (0.50)   1.52 (0.50)   0.953
  degree (mean (SD))         0.45 (0.50)   0.41 (0.49)   0.171
  nonwhite (mean (SD))       1.17 (0.38)   1.17 (0.37)   0.948
```

```r
# Run regressions and extract coefficients
balance_results_df <- map_dfr(covariates, function(covariate) {
  model <- lm(as.formula(paste(covariate, "~ treatment_fct")),
              data = data_subset)
  tidy(model) |>
    filter(term == "treatment_fct1") |>  # Modify if different factor levels
    mutate(covariate = covariate)
})

# Plot the coefficients for treatment effects on covariates
ggplot(balance_results_df, aes(x = estimate, y = covariate)) +
  geom_point(color = "blue") +
  geom_errorbarh(aes(xmin = estimate - 1.96 * std.error,
                     xmax = estimate + 1.96 * std.error), height = 0.2) +
  labs(title = "Effect of Treatment on Covariates",
       x = "Coefficient Estimate (Treatment Effect)",
       y = "Covariates") +
  geom_vline(xintercept = 0, linetype = "dashed", color = "red") +
  theme_minimal()
```



Exercise 4: Replicate the Main Finding by Varying Sample Size

Estimate the average effect of the treatment (ATE) on the outcome variable `support` conditional on the pre-treatment immigration attitudes `imm_1`. To do this, run the following script (change variable names if you made any modifications earlier):

```
# Run an OLS model with int. between treatment and immigration attitudes
## Change 'df' with the name you gave to the dataset
# model_study1 <- lm(support ~ treatment * imm_1, data = df)

# Check the results
# summary(model_study1)
```

The code line you just ran estimates an OLS regression with an interaction term between the experimental treatment and pre-treatment attitudes toward immigration, exactly as reported in the original paper. Compare these results with those of the authors.

Next, repeat the analysis using three progressively smaller random samples of the treatment and control groups (n = 200, n = 50, and n = 5 per group), resulting in total sample sizes of N = 400, 100, and 10, respectively.

Explain your findings: does the sample size affect the results? Why?

```r
# Set seed for replication
set.seed(123)

# Run an OLS model with int. between treatment and immigration attitudes
model_study1 <- lm(support ~ treatment_fct * imm_1, data = data_subset)
summary(model_study1)
```

```
Call:
lm(formula = support ~ treatment_fct * imm_1, data = data_subset)

Residuals:
    Min      1Q  Median      3Q     Max
-0.8926 -0.4344  0.1519  0.3038  0.7620

Coefficients:
                      Estimate Std. Error t value Pr(>|t|)
(Intercept)           0.238052   0.041717   5.706 1.47e-08 ***
treatment_fct1        0.148216   0.058093   2.551   0.0109 *
imm_1                 0.065453   0.006054  10.811  < 2e-16 ***
treatment_fct1:imm_1 -0.019267   0.008503  -2.266   0.0236 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4437 on 1147 degrees of freedom
Multiple R-squared:  0.134, Adjusted R-squared:  0.1317
F-statistic: 59.14 on 3 and 1147 DF,  p-value: < 2.2e-16
```

```r
# Set sample sizes for treatment and control groups
sample_sizes <- c(200, 50, 5)

# Function to sample and fit model
sample_and_fit <- function(n) {
  sample_data <- data_subset |>
    group_by(treatment_fct) |>
    sample_n(n) |>
    ungroup()

  # Fit the model on the sample data
  model_sample <- lm(support ~ treatment_fct * imm_1, data = sample_data)
```

```r
  # Summarize the model and return coefficients
  tidy(model_sample) |>
    filter(term == "treatment_fct1:imm_1") |> # Interaction term for CATE
    mutate(sample_size = n * 2) # Total sample size
}

# Apply function for each sample size and combine results
results_samples <- map_dfr(sample_sizes, sample_and_fit)
print(results_samples)
```

```
# A tibble: 3 x 6
  term                 estimate std.error statistic p.value sample_size
  <chr>                   <dbl>     <dbl>     <dbl>   <dbl>       <dbl>
1 treatment_fct1:imm_1  -0.0217    0.0146     -1.48   0.139         400
2 treatment_fct1:imm_1  -0.0340    0.0312     -1.09   0.278         100
3 treatment_fct1:imm_1   0.210     0.133       1.57   0.167          10
```

Submission guidelines

Please submit both the PDF file and the `.qmd` file. Both files should report all the code used for analysis and annotations explaining each step.

The name of the files must follow the structure *take-home_exercise_i_advanced_YOURSURNAME(S).pdf* and *take-home_exercise_i_advanced_YOURSURNAME(S).qmd*, respectively. They should be uploaded to the folder *Students responses/Take-home exercises/Take-home exercise I* in OLAT.

Deadline: **29.10.26**

References

Turnbull-Dugarte, S. J., & Ortega, A. L. (2024). Instrumentally inclusive: the political psychology of homonationalism. *American Political Science Review, 118(3)*, 1360-1378.